

Shubha Pramanik

AI/ML Developer | Frontend Developer | B.Tech CSE — 2nd Year

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PROFESSIONAL SUMMARY

Computer Science student with hands-on experience in AI/ML (RAG pipelines, NLP, content-based recommendation) and Full-Stack Web Development (React, Supabase, FastAPI). 1st Runner-Up at SIH Internal Hackathon (Team BlackPearl) and Winner at Hackaverse Hackathon. GSoC 2026 participant and active open source contributor. Building production-grade projects across AI/ML and Web domains, with a strong foundation in Python, JavaScript, and modern frameworks. Seeking internships in AI/ML, Frontend, or Backend Development.

TECHNICAL SKILLS

Languages C, Python, JavaScript, HTML5, CSS3	Core Concepts DSA, OOPs, RAG, NLP, Deep Learning, Generative AI, Content-Based Filtering, Problem Solving
AI / ML NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, PyTorch, LangChain, Hugging Face, Streamlit	Tools Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Netlify, Vercel
Web Dev React.js, Tailwind CSS, Bootstrap, Supabase, FastAPI	Deployment Streamlit Cloud, Vercel, Netlify

PROJECTS

■ RAG PDF Q&A; Assistant

LangChain | ChromaDB | Hugging Face Embeddings | PyMuPDF | RAG

AI-powered Retrieval-Augmented Generation system for intelligent, context-aware question answering over user-uploaded PDF documents.

- Semantic search with Hugging Face sentence embeddings and MMR retrieval for high-quality, non-redundant answers
- ChromaDB vector store for efficient document chunk indexing and cosine similarity lookup at query time
- End-to-end RAG pipeline: PDF parsing (PyMuPDF) → chunking → embedding → retrieval → LLM-based answer generation

■ InstaShare — Real-time Messaging Platform

React.js | Supabase | Real-time DB | JavaScript | Vercel

Production-grade instant messaging web app with real-time delivery, user authentication, and persistent chat history — built as a collaborative team project.

- Real-time bidirectional messaging via Supabase real-time database WebSocket subscriptions with zero-latency delivery
- Secure user authentication and persistent chat history managed entirely through Supabase backend
- Responsive React.js frontend deployed on Vercel for seamless cross-device accessibility

■ FlickPick — AI Movie Recommendation System

Python | Streamlit | FastAPI | Pandas | Scikit-learn | TF-IDF | TMDB API

Content-based AI movie recommendation engine using TF-IDF and cosine similarity with live TMDB API integration for personalised movie picks.

- TF-IDF vectorisation and cosine similarity over movie metadata for accurate content-based filtering
- Real-time movie posters and metadata fetched dynamically from TMDB REST API on every recommendation
- FastAPI backend decoupled from Streamlit frontend, enabling low-latency, scalable recommendation serving

EDUCATION

B.Tech — Computer Science Engineering	Siliguri Institute of Technology (SIT)	2nd Year, 3rd Sem 2024–2028
Class XII — WBCHSE	Balurghat L.M.A.U. Vidyalaya	411 / 500 (82.2%)
Class X — WBBSE	Bishrail High School	618 / 700 (88.3%)

ACHIEVEMENTS & ACTIVITIES

- 1st Runner-Up — SIH Internal Hackathon** (Team BlackPearl): Secured 1st runner-up position at the Smart India Hackathon internal round with project **Swachha Nagar**, a civic cleanliness and waste management solution.
- Winner — Beginner's Champion at Hackaverse Hackathon**: Awarded for outstanding innovation, execution, and problem-solving among beginner-track teams.
- Participant — Hack4Bengal Hackathon**: Competed in one of West Bengal's largest hackathons, delivering a functional product under tight time constraints.
- GSoC 2026 Participant** — Actively participated in Google Summer of Code 2026, engaging with open source organisations and the wider OS community.
- Open Source Enthusiast**: Regularly explores and contributes to open source repositories to sharpen real-world development skills.

- **Continuous Learner:** Self-studying advanced AI/ML (Deep Learning, NLP, Generative AI) and modern web technologies to remain industry-ready.

Open to internships in AI/ML | Frontend Development | Backend Development | Available immediately